

Deriving and Understanding the Black-Scholes Option Pricing Model

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Abstract

This paper explores the mathematical foundations of the Black-Scholes option pricing model in finance, a widely used model since its introduction in 1973. This paper will derive the Black-Scholes PDE and introduce the closed-form solution. Its closed-form solution allows for option prices to be computed quickly with relevant parameters. We examine the use of stochastic processes to model stock prices, utilizing key concepts such as Geometric Brownian Motion and Itô's Lemma to explain how volatility and randomness affect asset prices. This paper discusses the advantages, limitations, and assumptions of the Black-Scholes model, illustrating how these approaches can be utilized to calculate and manage financial risk in the financial market.

1 Introduction

The global stock market is estimated to be worth around \$127 trillion [2], with one source stating the average daily notional value of stocks traded is about \$405 billion a day. Options are a type of financial contract that contains leverage to buy or sell 100 shares of a stock at a fraction of the price of owning a stock. Starting in 2021, the average daily nominal value of options traded reached \$450 billion a day, higher than stock trading at that time. Before the Black-Scholes formula was introduced in 1973, options were traded largely on rules of thumb and intuition. The introduction of the Black-Scholes model, developed by Fischer Black, Myron Scholes, and Robert Merton, provided a mathematical foundation for pricing options and allowed for the growth of the multi-trillion-dollar options industry.

The key concept that Black-Scholes introduced was using dynamic hedging and risk-neutral pricing to find the theoretical price of an option based on variables such as stock price, strike price, time to expiration, volatility, and risk-free rate.

The value of options depends on the future price of the underlying stock. This creates a challenge where it can be difficult to price options since the stock price is uncertain. There are many events influencing the stock market and individual stocks, making their movements appear random at times. Stochastic calculus provides the mathematical framework for analyzing financial markets and random price movements. Without it, typical calculus is not enough to manage the non-differentiable processes that model asset prices. The Black-Scholes model views stocks as Geometric Brownian Motion (GBM), a type of stochastic process, since both exhibit similar random movements. An essential stochastic tool is Itô's Lemma, which allow us to differentiate the option price function, which depends on the stochastic GBM stock process.

The properties and types of stochastic processes are essential tools in the Black-Scholes equation and PDE, so this paper will define the stochastic tools used in the Background. Additionally, financial definitions and examples will be given. In the final section, we will use Itô's Lemma to derive the Black-Scholes PDE and introduce the closed-form solution.

2 Background

This section establishes the foundational financial and mathematic definitions and concepts throughout the paper. We begin by introducing options and their role in reducing risk. Then, we introduce the stochastic tools forming the mathematical foundation of the Black-Scholes model.

2.1 Financial Background

Definition 2.1. An *option* is a security that gives the right, but not the obligation, to buy or sell an asset at a set price within a specific period of time.

Options are important tools for controlling risk in portfolios, as they allow investors to limit losses at the cost of the initial fee paid (the premium). For example, if the underlying asset price goes down to 50% of the original price, an asset holder would lose that amount of money while a *call option* holder would choose not to exercise and only lose the premium.

Definition 2.2. A *call option* is a type of option that gives the right to buy an asset at a set price within a specific period of time.

This kind of option profits when a stock price rises above the *strike price* plus the premium paid.

Definition 2.3. The *strike price* is the specified price for which an option-holder buys or sells an asset when exercising an option.



Figure 1: Profit/Loss Graph of a Call Option

Notice how the graph shows potentially unlimited gains and losses capped at the premium paid. This contrasts with how owning a stock can have up to a 100% lost. Options can be viewed as a reduction of risk at the cost of a small fee.

There are also multiple kinds of options based on when they can be exercised, such as American options, which can be exercised at any point until the expiration date. Black-Scholes gives a closed-form solution to European options specifically. American options do not have a closed-form solution and involve methods such as solving an option's pricing PDE or simulating thousands of potential stock paths.

Definition 2.4. A *European option* is a type of option that can only be exercised only on a specific future date.

Example 2.5. Consider a single share of Apple with an initial stock price of \$100. If we buy a call option at a premium of \$5 with a strike price (the price we want the option to buy at) of \$110, which expires in a year. The option makes a profit if the stock goes above \$115 (the break-even price), and can only lose a maximum of \$5.

Definition 2.6. The *risk-free rate* is the rate of return of an asset that can be invested in without risk, usually based on government bills/bonds that essentially have no risk of default or loss.

Since we have a risk-free rate to invest capital into, the value of future profits/costs should be adjusted to today's value. This is done with a *discount factor*.

Definition 2.7. Given a constant risk-free rate r , the *discount factor* is $e^{-r(T-t)}$, for current time t and future time T . We use the discount factor to convert future value into today's value because it shows how much of today's money would have to be invested at the risk-free rate to gain the same amount in the future.

Accounting for the time value of money invested without risk is important for assessing future value and present cost.

Example 2.8. Given current time $t=0$, future time $T = 1$, constant risk-free rate $r = 0.05$, and future payment of \$100, we can calculate the discount factor with $e^{-0.05(1-0)} = e^{-0.05} \approx 0.951$. Calculating the present value:

$$\text{Present Value} = e^{-0.05(1-0)} * \text{Future Payment} \approx 0.951 * \$100 = \$95.1$$

We interpret this as a payment of \$100 in 1 year is worth \$95.1 today if invested at a risk-free rate of 5%. If an asset pays \$100 in 1 year, its current price should be \$95.1. However, the future payment of an asset or option is unknown as a random variable, so we have to use the expected value of the future payment.

Definition 2.9. *Shorting* is an investment strategy where an investor borrows an asset, sells it on the market, and buys the same asset later on. The profit (or loss) is the difference between the price the borrowed stock was sold at and the price at which it was bought again.

When the stock that an investor is shorting increases in price, the investor loses money, while if the stock goes down, the investor profits. Shorting is used to *hedge* against risk in a portfolio since it moves opposite of the underlying stock.

Definition 2.10. *Hedging* is a practice that reduces or eliminates risk in a *portfolio* (a selection of assets) by choosing its composition such that random changes in one asset is offset by opposing fluctuations in another. *Dynamic hedging* involves continuously adjusting a portfolio based on market movements to maintain a specific risk exposure, such as aiming for zero risk.

Shorting the underlying stock is used in the Black-Scholes model to dynamically hedge against price fluctuations of the option, effectively eliminating theoretical risk in the portfolio.

2.2 Stochastic Background

A *stochastic process* is a process that can be described with the change of random variables over time. An important kind of stochastic process used when modeling phenomenon with random “noise” or fluctuations is known as a *Brownian motion/Wiener process*.

Definition 2.11. Let B_t be a stochastic process. B_t is a *Brownian Motion/Wiener process* if it satisfies the following for $t \geq 0$:

1. $B_0 = 0$.
2. B_t is continuous.
3. Each future increment (difference in value) is independent of the past increments. For $u > 0$, $B_{t+u} - B_t$ is independent from past values $B_s, s < t$.
4. For all $t \geq 0, u \geq 0$, the increment $B_{t+u} - B_t$ is represented by a normal distribution, $N(0, u)$

These properties of Brownian motion make it especially useful for modeling the random behavior for stocks in finance. Brownian motion exhibits similar random fluctuations in stocks. A type of stochastic process that makes use of Brownian motion is *Itô Processes*.

Definition 2.12. An *Itô Process* is a stochastic process represented by

$$dX_t = \mu(t, X_t) dt + \sigma(t, X_t) dB_t$$

where

- B_t is a standard Brownian motion
- $\mu(t, X_t)$ is the drift coefficient (deterministic part)
- $\sigma(t, X_t)$ is the diffusion coefficient (stochastic part)

Itô processes models random systems evolving over continuous time and allow for changing drift and volatility. However, *GBMs* are a type of Itô process with constant drift (average return) and constant volatility (day-to-day deviations in return). GBMs can be used to model stocks with their average return as the drift and calculated volatility over a period of time using historical data. The Black-Scholes model's use of constant variance and drift simplifies the complex problem of pricing options. While an unrealistic assumption, the model still provided a useful framework to price options and manage risk.

Definition 2.13. A stochastic process S_t is a *Geometric Brownian Motion (GBM)* if it satisfies the stochastic differential equation (SDE):

$$dS_t = \mu S_t dt + \sigma S_t dB_t,$$

Where B_t is a Brownian motion, and μ is the drift (in percentage) and σ is the volatility (in percentage).

An analytical solution to the stochastic differential equation above is

$$S_t = S_0 e^{\left(\mu - \frac{\sigma^2}{2}\right)t + \sigma B_t}. \quad (1)$$

GBMs are used to model stock prices because GBMs don't allow negative values/prices. Brownian motion on its own doesn't have drift and can become negative. GBMs also display similar “jaggedness” as real stock prices, and the expected growth of GBMs is independent of the current stock price, similar to real stocks.

Additionally, real-life stock returns are historically log-normal with daily moves in percentage value, rather than absolute dollar terms. For example, a \$10 increase in a \$100 stock is much different than a \$10 increase in a \$10 stock. It makes more sense to view stock price movements in terms of percentage increases from the previous time period, such as a stock being down 10% from yesterday.

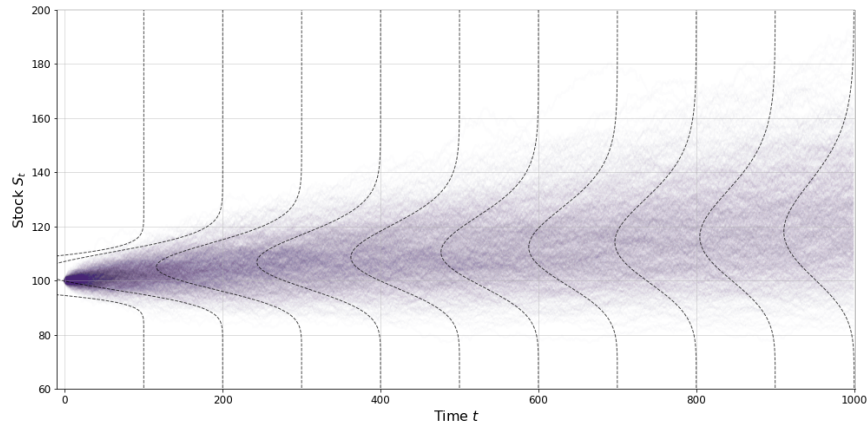


Figure 2: Geometric Brownian Motion with Slight Upward Drift

However, in real stock prices, volatility changes over time, whereas GBMs assume constant volatility. Major news or events also cause real prices to jump rapidly, which GBMs can't model as they are continuous.

3 Black-Scholes Derivation

This section will go over the derivation of the Black-Scholes PDE and introduce the closed-form solution, using the financial and stochastic tools introduced in the Background section. The use of Itô's Lemma is important in proving the Black-Scholes PDE since stochastic processes cannot be differentiated in the same way as non-stochastic processes. Itô's Lemma allows us to find the derivative of a stochastic process that depends on time t and another stochastic process X_t , which we will set as our GBM for stock price, S_t

Lemma 3.1. (Itô's Lemma) *Let $f(t, x)$ be a twice differentiable function of two variables, and let X_t be a stochastic process satisfying $dX_t = \mu_t dt + \sigma_t dB_t$ for a Brownian motion B_t . Then,*

$$df(t, X_t) = \left(\frac{\partial f}{\partial t} + \mu_t \frac{\partial f}{\partial x} + \frac{1}{2} \sigma_t^2 \frac{\partial^2 f}{\partial x^2} \right) dt + \sigma_t \frac{\partial f}{\partial x} dB_t.$$

Essentially, this is analogous to the chain rule in deterministic calculus. Itô's Lemma is important for finding a solution to the Black-Scholes equation because it allows us to compute the derivative of a stochastic process, such as one that models option prices. Since options pricing is based on how much we expect a stock price to go up (or down) within the specified time frame, the derivative of an option price gives us an expression for how the option's value changes with respect to time and the underlying stock price.

Assumptions for the Black-Scholes model:

- Stock price follows Geometric Brownian Motion: $dS_t = \mu S_t dt + \sigma S_t B_t$ for Brownian motion B_t and constant μ and σ . This means that prices are continuous, returns are independent over time, and returns are log-normally distributed.
- Interest rates and risk-free rate r are constant.
- The option is European style.
- No-arbitrage market and frictionless transactions, meaning no risk-free profit and no transaction costs/taxes/delays in the market.

Theorem 3.2. *Consider a European option with price $V(t, S)$, where S is the underlying stock price satisfying the geometric Brownian motion*

$$dS = \mu S dt + \sigma S dB_t.$$

Using Itô's Lemma for $V(t, S)$ and hedging arguments, the Black-Scholes partial differential equation is

$$\frac{\partial V}{\partial t} + rS \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} = rV,$$

where r is the risk-free interest rate and σ is the volatility of the stock.

Proof. Let S_t be the price of a stock at time t , following a Geometric Brownian Motion. Thus,

$$dS = \mu S dt + \sigma S dB_t. \quad (2)$$

Let $V(t, S)$ be the price of a European option as a function of the stock's price and time. Applying Itô's Lemma 3.1 on $V(t, S)$ gives:

$$dV = \left(\frac{\partial V}{\partial t} + \mu_t \frac{\partial V}{\partial S} + \frac{1}{2} \sigma_t^2 \frac{\partial^2 V}{\partial S^2} \right) dt + \sigma_t \frac{\partial V}{\partial S} dB_t \quad (3)$$

From equation (2), we find the drift term is $\mu_t = \mu S$ and the stochastic term $\sigma_t = \sigma S$. Substituting both into (3) yields:

$$dV = \left(\frac{\partial V}{\partial t} + \mu S \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \right) dt + \sigma S \frac{\partial V}{\partial S} dB_t \quad (4)$$

Then, to eliminate the stochastic term B_t , we form a portfolio with one option V and *shorting* Δ stocks of S to perform *dynamic hedging*:

$$\Pi = V - \Delta S \quad (5)$$

Taking the derivative and substituting dS gives

$$d\Pi = \left(\frac{\partial V}{\partial t} + \mu S \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \right) dt + \sigma S \frac{\partial V}{\partial S} dB_t - \Delta(\mu S dt + \sigma S dB_t) \quad (6)$$

$$= \left(\frac{\partial V}{\partial t} + \mu S \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \right) dt - \Delta \mu S dt + \left(\sigma S \frac{\partial V}{\partial S} - \sigma \Delta S \right) dB_t \quad (7)$$

Choose $\Delta = \frac{\partial V}{\partial S}$ to cancel out the dB_t term in equation (7). This removes the effect of Brownian motion in the hedged portfolio Π , removing risk and leaving only the drift term for $d\Delta$. By the no-arbitrage principle, since the portfolio is *locally risk-free* (no randomness/risk-free over an infinitesimally small time interval), this portfolio must earn the risk-free rate. Therefore, with $\Delta = \frac{\partial V}{\partial S}$:

$$d\Pi = r\Pi dt \quad (8)$$

$$= r(V - S \frac{\partial V}{\partial S}) dt = \left(rV - rS \frac{\partial V}{\partial S} \right) dt \quad (9)$$

$$= \left(\frac{\partial V}{\partial t} + \mu S \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \right) dt - \frac{\partial V}{\partial S} \mu S dt \quad (\text{From equation (7)}) \quad (10)$$

$$= \left(\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \right) dt \quad (11)$$

Equating (9) and (11) and moving term to the opposite side of rV yields:

$$rV = \frac{\partial V}{\partial t} + rS \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \quad (12)$$

□

Thus, we have shown how the Black-Scholes PDE equation is derived. We can see from the PDE that the price of an option depends on time, the risk-free drift of the stock S , and the volatility σ of the underlying stock.

The term $\frac{\partial V}{\partial t}$ is the time-value of an option. The more time until the expiration date, the more time available for favorable stock movements to occur. As time goes on, the price of an option tends to decrease if the stock price is constant. This is because less time to expiration leads to a more predictable stock price at expiration.

The term $rS \frac{\partial V}{\partial S}$ represents the risk-free return of the portfolio consisting of $\frac{\partial V}{\partial S} = \Delta$ stocks, using the same Δ from the proof above. As the hedged portfolio Π grows at the risk-free rate, the price of the option also increases by the rate $r\Delta S$.

The third term $\frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2}$ represents the contribution of stock volatility in the price of an option. The profit/loss of an option is asymmetric, with capped losses and potentially unlimited gains. This means that higher volatility increases the likelihood of an extreme payout, benefiting the option holder.

However, we can improve upon this by finding a closed-form formula rather than having to solve a PDE to price an option. This helps save computational power and time, which is essential in the fast-paced market where milliseconds can lose profit or lose an advantage in High-Frequency Trading. To find the closed-form solution, we rely on *risk-neutral measure* and the concept of *martingales*.

Definition 3.3. A *martingale* is a stochastic process that, given all the past information, has the expected future value equal to its current value. That is, there is no expected gain or loss, so it is a “fair game.” Formally, for a martingale X ,

$$\mathbb{E}(X_T) = X_t, T > t.$$

Additionally, a martingale process has no drift term, only the stochastic term with B_t .

Martingales are a useful concept in the Black-Scholes model since using *risk-neutral measure and pricing* turns the discounted asset price into martingales, allowing us to state that the drift term of a stock is equal to the risk-free rate under *risk-neutral measure*.

Definition 3.4. *Risk-Neutral Measure* $\mathbb{Q}$ is a probability measure in which the expected return of a stock is r under $\mathbb{Q}$. This makes the expected future discounted price of an asset equal to its current value. Thus, the discounted asset price is a martingale.

Risk-Neutral Pricing under $\mathbb{Q}$ of a call option with:

- $V(t, S)$ = Price of call at time t and stock price S
- S = Underlying stock price
- $\tau = T - t$ = Time to maturity
- r = Risk-free rate
- $\mathbb{E}^{\mathbb{Q}}$ = Expected value under risk-neutral measure $\mathbb{Q}$
- $\max(S_T - K, 0) = (S_T - K)^+$ Payoff function at maturity
- $\mathcal{F}_t$ = Information available at time t (known as filtration)

$$V(t, S) = e^{-r\tau} \mathbb{E}^{\mathbb{Q}}[(S_T - K, 0)^+ | \mathcal{F}_t] \quad (13)$$

In other words, the fair price of an option under $\mathbb{Q}$ is the present value of its expected payoff, given all that you know currently. Moving to the risk-neutral world allows us to state that the expected return of the stock is equal to the risk-free rate r , eliminating the need to consider trader preferences and risk aversion. The usage of risk-neutral measure in the Black-Scholes model simplifies the problem of pricing options and other financial derivatives.

The proof for the closed-form Black-Scholes formula using risk-neutral measure is shown by Younggeun Yoo [3]. Alternatively, the Black-Scholes PDE can be transformed into the standard heat PDE from physics, which has a known solution and can be solved with the boundary condition $V(T, S_T) = \max(S_T - K, 0)$.

Theorem 3.5. *The closed-form formula for the price of a European call option by Fischer Black and Myron Scholes [1] is*

$$C_0 = S_0 N(d_1) - K e^{-rT} N(d_2),$$

where

$$d_1 = \frac{\ln\left(\frac{S_0}{K}\right) + (r + \frac{\sigma^2}{2})T}{\sigma\sqrt{T}}, d_2 = \frac{\ln\left(\frac{S_0}{K}\right) + (r - \frac{\sigma^2}{2})T}{\sigma\sqrt{T}} = d_1 - \sigma\sqrt{T},$$

and $N(x)$ is the cumulative distribution function for a normally distributed variable x . T is time to maturity (in years), r is the risk-free rate (as a decimal/percentage), S_0 is the starting stock price, σ is the annualized standard deviation of stock returns, and K is the strike price.

The first term $S_0 N(d_1)$ represents the option's current value as owning Δ stocks in the hedged portfolio created in the Black-Scholes proof. This is because $N(d_1) = \Delta$ from the proof.

The second term $K e^{-rT} N(d_2)$ represents the present cost of exercising the option and buying the stock at the price K , multiplied by the probability that the option is profitable, which is $N(d_2)$.

The formula above applies to European call options, but for European put options, which allow the option holder to sell a stock at a strike price, the formula is analogous, with some adjustments. The formula focuses instead on the right to sell the asset at the strike price rather than focusing on the potential gain from exercising a call option. The put option formula by Black-Scholes [1] is very similar.

$$P_0 = K e^{-rT} N(-d_2) - S_0 N(-d_1),$$

where

$$d_1 = \frac{\ln\left(\frac{S_0}{K}\right) + (r + \frac{\sigma^2}{2})T}{\sigma\sqrt{T}}, d_2 = \frac{\ln\left(\frac{S_0}{K}\right) + (r - \frac{\sigma^2}{2})T}{\sigma\sqrt{T}} = d_1 - \sigma\sqrt{T}$$

Example 3.6. Consider a stock that begins at \$100, 1 year to expiration ($T=1$), risk-free interest rate $r = 4\%$, standard deviation of stock returns is 20% , and strike price is \$110.

Using the closed-form solution above, we have

$$d_1 = \frac{\ln(\frac{100}{110}) + (0.04 + 0.5 * (0.20)^2) * 1}{0.20 * 1} = -0.17655$$

$$d_2 = \frac{\ln(\frac{100}{110}) + (0.04 - 0.5 * (0.20)^2) * 1}{0.20 * 1} = -0.37655$$

$$C_0 = 100 * N(-0.17655) - 110e^{-0.04*1}N(-0.37655) = 5.65084$$

So we would price the value of the option at \$5.65084. If someone is selling below what we believe the option is worth, that gives some indication that buying that option is a good investment. Similar to trading stocks, we want to buy low and sell high to make a profit.

However, the Black-Scholes model is limited in some ways. This includes the assumptions of constant drift and volatility. In reality, volatility and stock growth can fluctuate with global events, earnings reports, and sudden shocks in the market, which can lead to mispricing. The risk-free rate r may also change over time. Additionally, the formula does not take into account dividend payments, which incentivize exercising, and companies paying out dividends tends to decrease the stock price. However, there are variations that have been adapted to consider this, such as the Black-Scholes-Merton extension created by Robert Merton. Models such as the Heston model view volatility as its own stochastic process rather than a constant, allowing for a more accurate price. Other models also allow for price jumps seen in stocks, which cannot be modeled by continuous GBMs.

Finally, the model is restricted to European options and is unable to accurately price the more complex American options, which Black-Scholes tends to underprice. American options can be exercised at any time before the expiration date and thus require more complex models, such as binomial tree models.

Despite its limitations, the Black-Scholes remains a cornerstone in quantitative finance. Many modern models build on its discoveries, extending it to more accurately predict the real market.

References

- [1] Fischer Black and Myron Scholes. The pricing of options and corporate liabilities. *Journal of Political Economy*, 81(3):637–654, 1973. URL: <http://www.jstor.org/stable/1831029>.
- [2] Jenna Ross. The \$127 Trillion Global Stock Market in One Giant Chart. *VisualCapitalist*, Aug 2025. <https://www.visualcapitalist.com/sp/ter01-piecing-together-the-127-trillion-global-stock-market>.
- [3] Younggeun Yoo. Stochastic Calculus and Black-Scholes Model. *University of Chicago, Mathematics REU*, 2017. URL: <https://math.uchicago.edu/~may/REU2017/REUPapers/Yoo.pdf>.